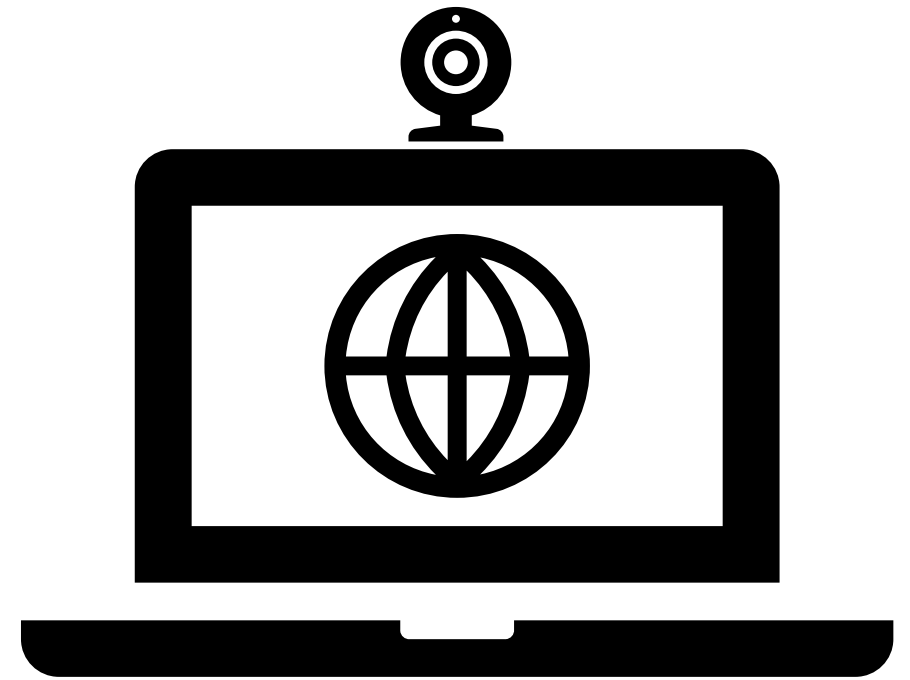


ONLINE EXPERIMENTS

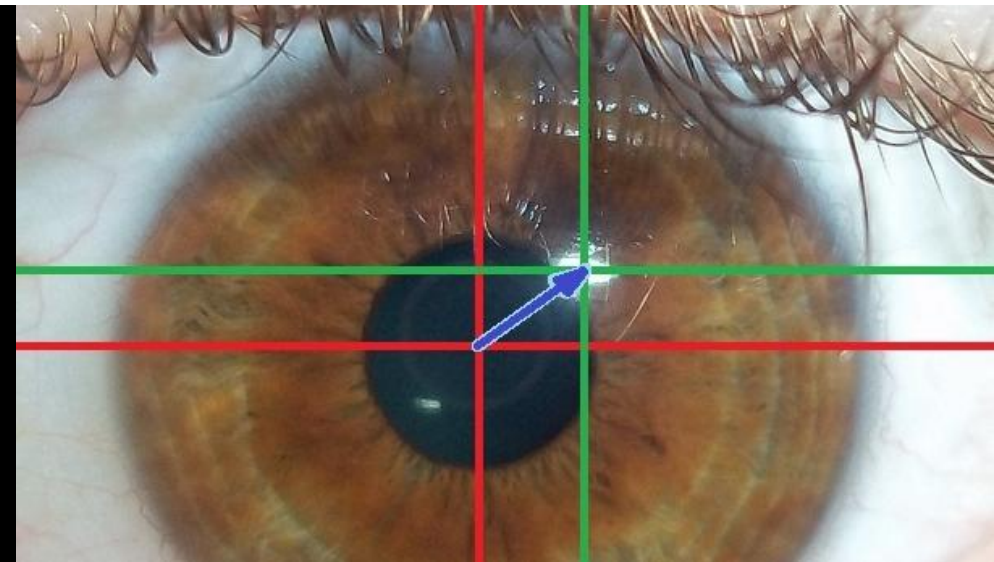


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Research Data & Methods

1 October 2025, Uni Köln

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ROADMAP



Motivation



Concept



Limitations



Step-by-step



Analysis & Results

ROADMAP



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Analysis & Results



PANDEMIC

Non-WEIRD populations

(Western Educated Industrialized Rich Democratic, Henrich, 2020)

Regions of the world with no laboratories nearby

Populations that prefer spaces outside of university

- Heritage speakers of Turkish in Germany



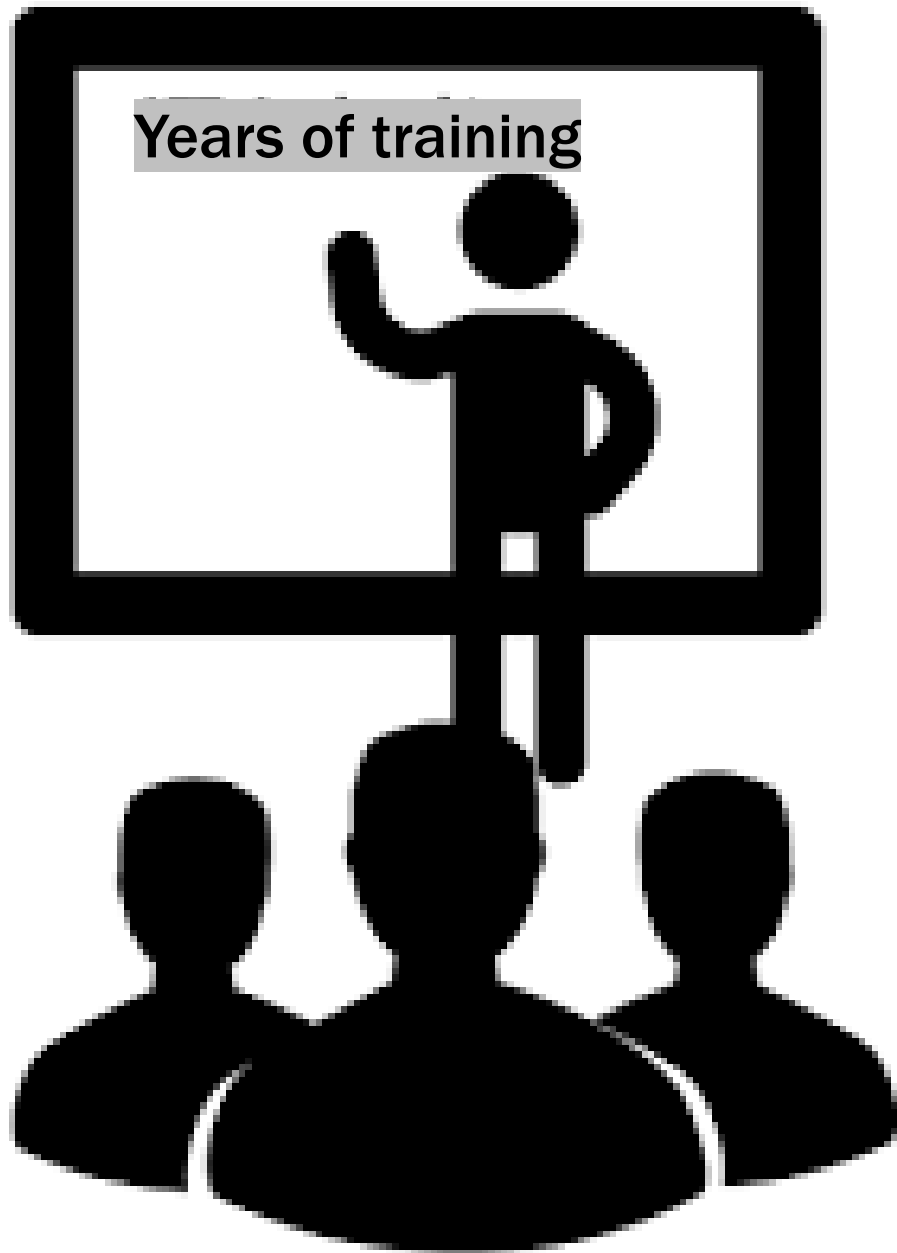
REMOTE AND HARD-TO-REACH POPULATIONS

High-end eye-trackers
\$10,001+

Webcam-based eye-tracker
No additional cost

PRICE / COST





TECHNICAL TRAINING

Running webcam-based eye-tracking (ET) experiments is possibly more approachable than the lab-based method.

Online training, online support, ...

ROADMAP



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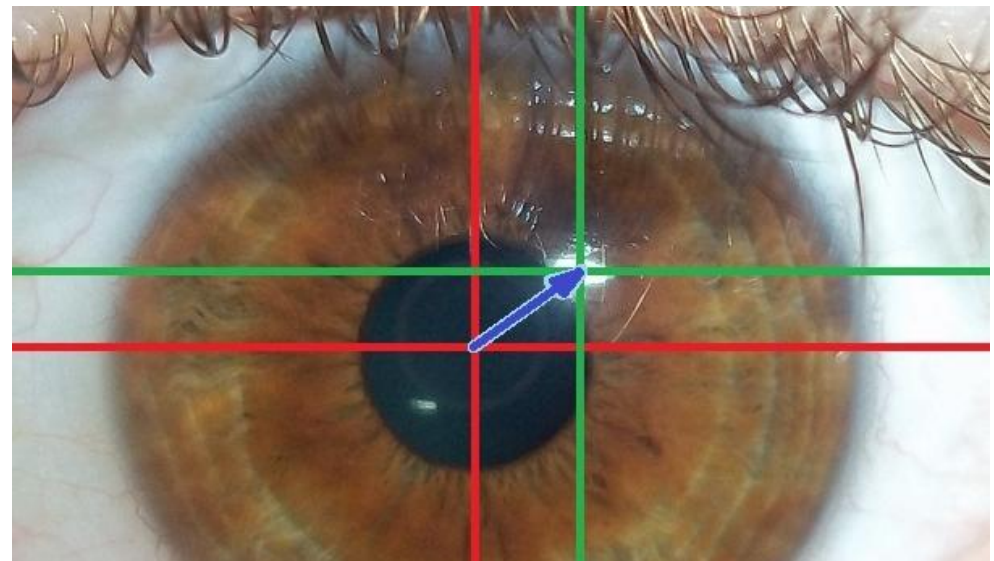
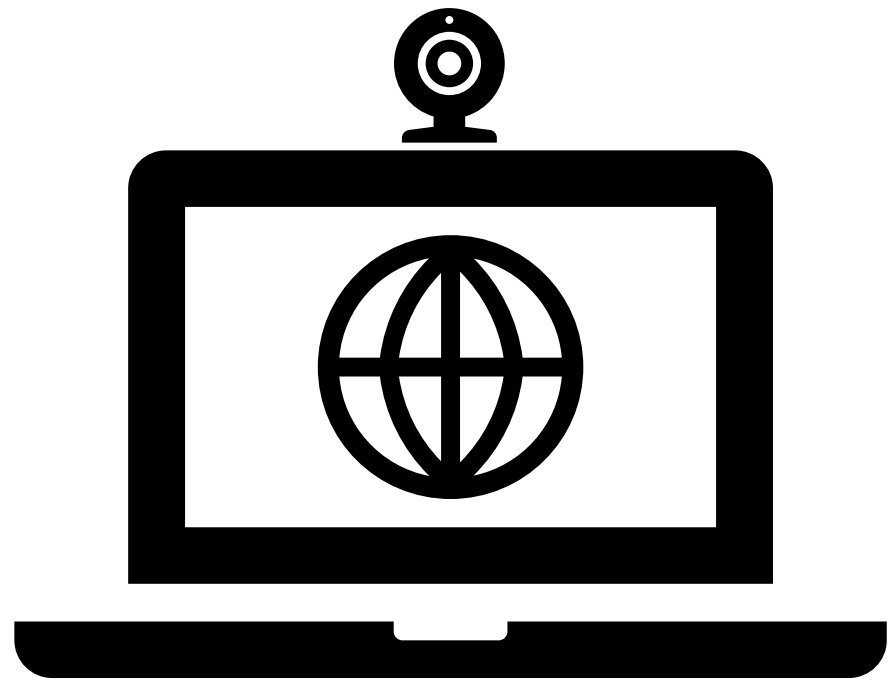


Step-by-step



Analysis & Results

WEBCAM TRACKS MOVEMENT OF THE PUPIL



WEBGAZER.JS

"Democratizing Webcam Eye Tracking on the Browser"

<https://webgazer.cs.brown.edu/>

EYE-TRACKING COORDINATES ARE COMPUTED IN PARTICIPANTS' BROWSER

```
...object to mirror_mod.mirror_object
operation == "MIRROR_X":
    mirror_mod.use_x = True
    mirror_mod.use_y = False
    mirror_mod.use_z = False
operation == "MIRROR_Y":
    mirror_mod.use_x = False
    mirror_mod.use_y = True
    mirror_mod.use_z = False
operation == "MIRROR_Z":
    mirror_mod.use_x = False
    mirror_mod.use_y = False
    mirror_mod.use_z = True
```

```
...selection at the end -add
...selected = 1
...selected = 1
context.scene.objects.active
("Selected" at the end)
mirror_ob.select = 0
= bpy.context.selected_object
data.objects[one.name].select
print("please select exactly
```

-- OPERATOR CLASSES -----

```
types.Operator):
    X mirror to the selected
    object.mirror_mirror_x"
    mirror X"
```

```
...):
    ... object is not
```



PCibex

One of several platforms to program and host webcam-based ET experiments

Free

Great support via forum and Slack

Detailed documentation: <https://doc.pcibex.net/>

Zehr, J., & Schwarz, F. (2018). PennController for Internet Based Experiments (IBEX).

<https://doi.org/10.17605/OSF.IO/MD832>



PCibex

Types of experiments in PCibex:

Surveys, questionnaires, acceptability judgement tasks, rating tasks, stroop tasks, picture selection tasks, audio recordings, eye-tracking, lexical decision tasks, mouse tracking, ...

PCIBEX



Really good documentation
with tutorials!

placements

concepts

tutorial

ed Tutorial

guides

ts

nds

t

ntroller 2.0

rs

Search PCIBex Documentation

Go to the PCIBex Farm

PennController for IBEX

A free, open-source, versatile, and user-friendly online experiment builder.

TRY A DEMO

GET STARTED

TAKE THE TUTORIAL

About

PennController for Internet Based Experiments ("PennController" or "PCIBex" for short) provides the tools to build and run online experiments, from familiar paradigms like self-paced reading to completely custom-designed paradigms.

The **PCIBex Farm**'s interface comes with [many features](#) to help you design and run your experiments, including:

ROADMAP



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Step-by-step



Analysis & Results

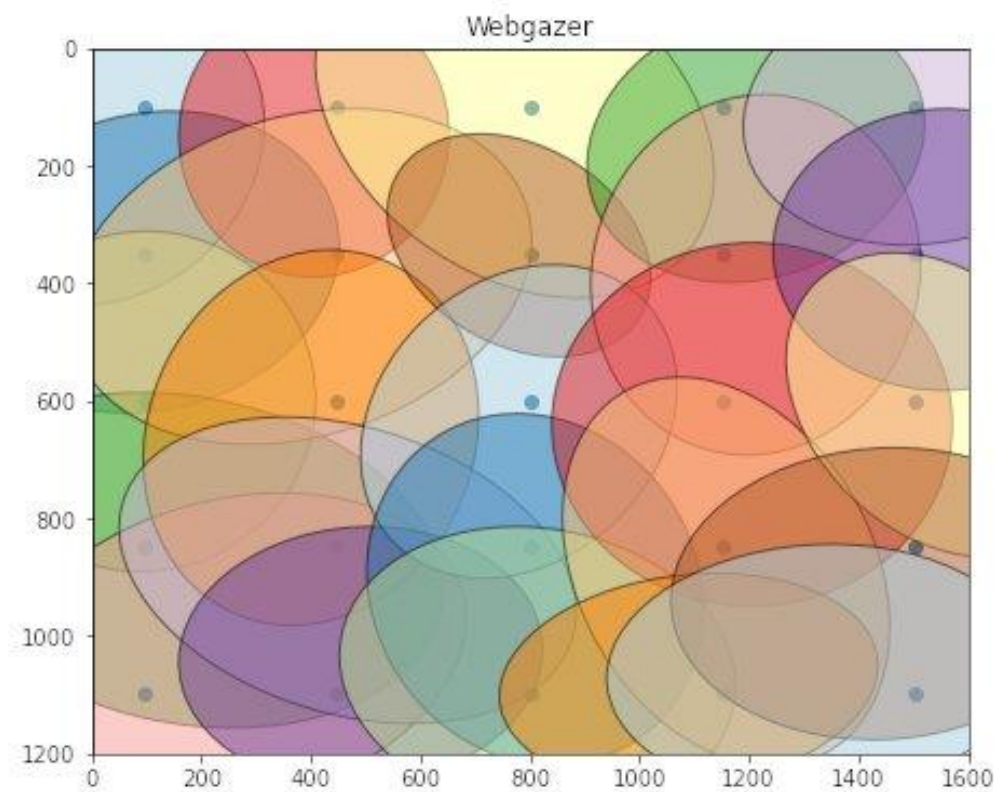
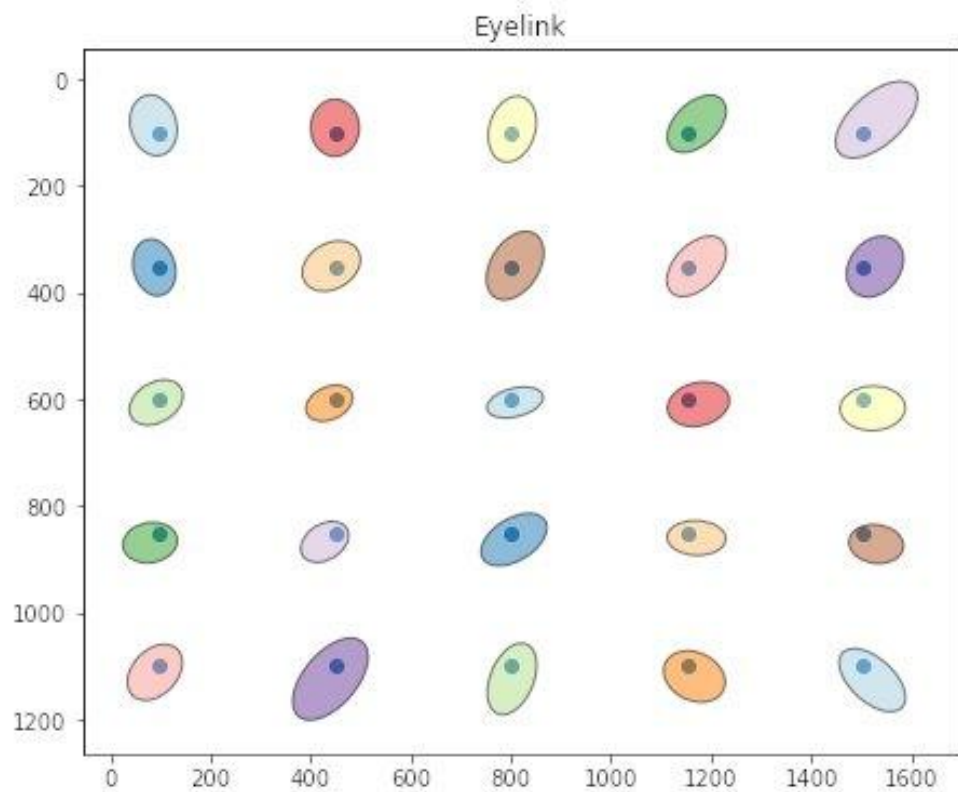


DELAYS

On most trials, there are delays and variation between the actual time point of an eye gaze and the recorded time.

~300ms on average
(Slim & Hartsuiker, 2022)

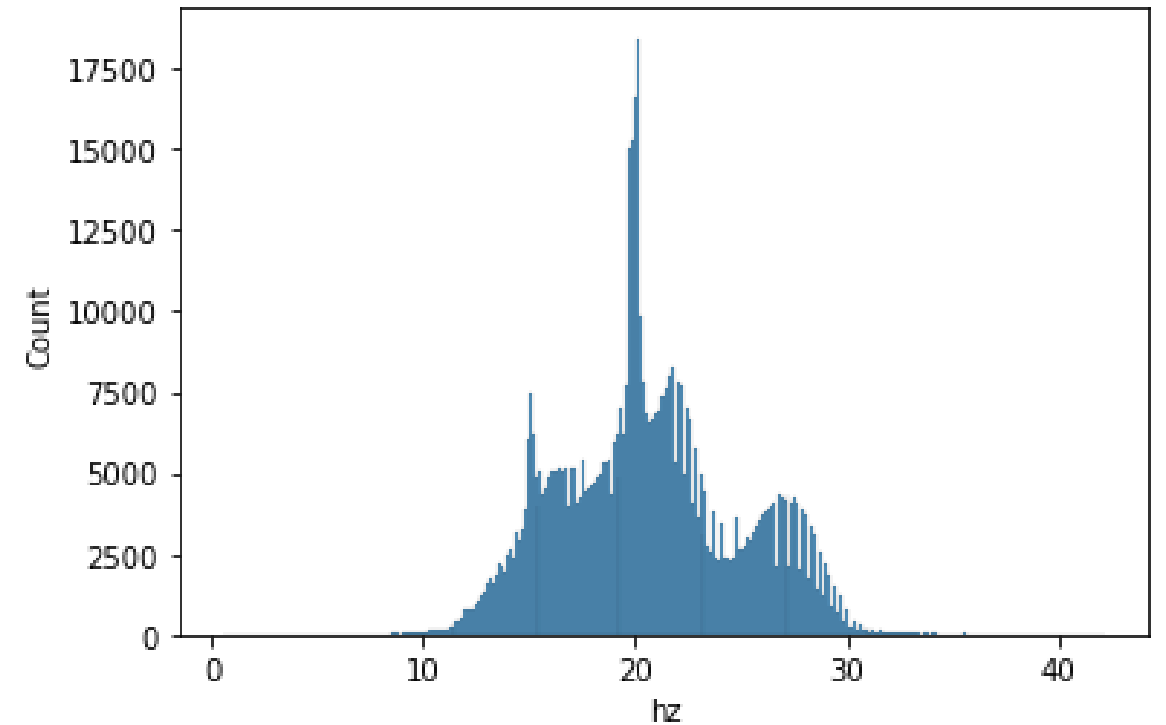
ACCURACY



(Zhang et al., 2022, [Tweet](#))

VARYING SAMPLING RATE

(Zhang et al., 2022, [Tweet](#))



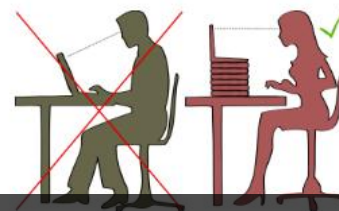


It looks like we were not able to precisely calibrate the tracker:

You calibration score is 48 and you need at least 50

Here are a few tips to help you better self-calibrate:

- try adjusting your webcam based on the video in the top-left corner.
- if you use an external webcam, make sure it is fixed to the top of your screen.
- try raising your screen so as to align your webcam with your eyes



DIFFICULT TO CALIBRATE

- make sure you are not wearing eyeglasses reflecting ambient light.

- make sure the algorithm detects your face (it should appear green).

- make sure there is enough ambient light for face-detection.

- make sure you follow your mouse pointer with your eyes.

ROADMAP



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Analysis & Results

CREATING AN ACCOUNT

Go to <https://farm.pcibex.net/> and click "Sign Up" and create an account with a username, password and email address.

You can create as many accounts as you want (and you may have to).

STEP-BY-STEP GUIDELINES FOR AN EYE-TRACKING EXPERIMENT

<https://hu.berlin/ET>



HU BERLIN FARM

<https://korpling.german.hu-berlin.de/ibex/>

FIRST STEPS IN BUILDING AN EXPERIMENT

We will review the code / script of an experiment together.

It has three parts:

Translation tasks

Stroop task

Survey

FIRST STEPS IN BUILDING AN EXPERIMENT

Go to this demo link and copy the experiment into your account:

<https://farm.pcibex.net/r/jsBNEF/>



FIRST STEPS IN RUNNING AN EXPERIMENT

Go to this demo link and do the study :

<https://farm.pcibex.net/p/XmseQ0/>

ROADMAP



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Concept



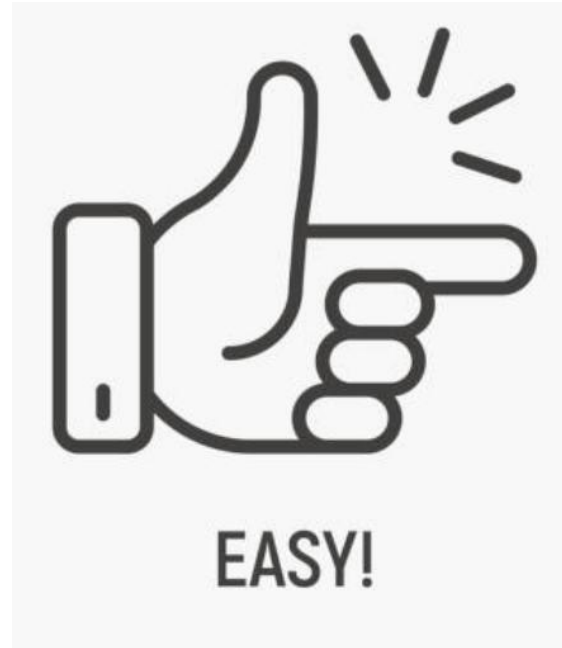
Limitations



Step-by-step



Analysis & Results



JUMP TO R

**PCibex / Webcam-based eye-tracking
offers easily accesible and free method**

"Good enough" for VWP

Serious limitations to consider in advance

**Consider "supervised" webcam-based ET
to improve data quality and control**

...



OTHER EXPERIMENTS AND ANALYSIS SCRIPTS

Analysis Turkish case processing:

https://osf.io/sehnf/?view_only=a0a306606f064652b14b03e29a52d27d

Turkish case processing experiment:

<https://farm.pcibex.net/r/GOsYzn/>

German case processing experiment:

<https://farm.pcibex.net/r/gcwFgV/>